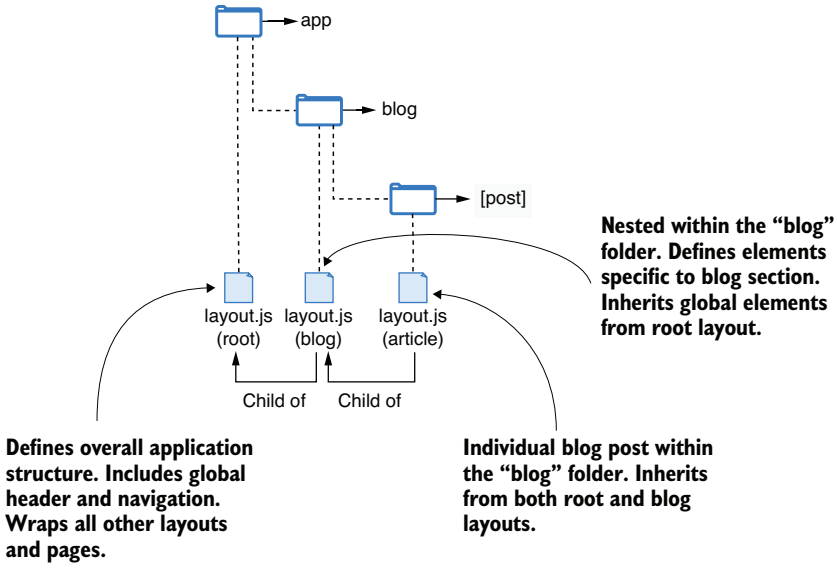


Figure 2.10 This diagram shows nested layouts in Next.js. The root layout wraps all content, while the blog layout adds elements specific to the blog section. Individual blog posts are inherited from both.

Every Next.js application requires a single root layout. This layout, typically defined in a layout.js file at the top level of the app directory, serves as the foundation for your application's UI. It's crucial because it's the only layout component that can include the mandatory `<html>` and `<body>` tags. The code in the following listing shows the minimal layout.js used in our application for reference.

Listing 2.6 Root layout component

```
import './index.css';
export const metadata = {
  title: 'Astra AI',
  description: "Hello, I'm ✳ Astra. Ask me anything you want.",
};

export default function RootLayout({ children }) {
  return (
    <html lang="en">
      <body>{children}</body>
    </html>
  );
}
```

Defines application metadata (used for SEO and rendering)

Root layout component

Renders all the children's components based on the defined routes

The `children` prop in the `RootLayout` component plays an important role. It acts as a placeholder for all the components that Next.js renders based on your application's